

LEARNED NYQUIST: FUNDAMENTAL ALIASING LIMITS OF IMPLICIT NEURAL REPRESENTATIONS

Weiping Yan

University of Minnesota, Twin Cities, Minneapolis, MN, USA

ABSTRACT

Implicit neural representations (INRs)—SIREN and Fourier-feature coordinate networks—are increasingly used as continuous models of signals. A recent *achievability* sampling theory addresses signals that are themselves INRs; we address the *converse* question that governs practice, where the signal is arbitrary and observed at finite, noisy, nonuniform samples. A fixed-feature INR (fixed feature frequencies, linear output layer) represents only a *structured* frequency set Λ . Casting its least-squares fit as generalized sampling on this dictionary, we prove: (i) achievability—if the signal spectrum lies in Λ and the samples render the dictionary matrix full column rank, a bandwidth-matched INR recovers it with closed-form noise gain $\kappa = 1/\sigma_{\min}(\Phi)$; (ii) a converse—out-of-band energy is *folded* onto Λ with an error floor no fixed-feature INR can beat, and for grid-coherent tones the sample residual is *exactly zero* while the reconstruction error is at least the full out-of-band energy (*silent aliasing*), with a *non-classical* fold target when Λ is structured; (iii) an exact decomposition of the estimator’s average error into an estimation-variance term and an aliasing-variance term that no sampling density uniformly removes. We give a bandwidth-matched estimator and a ground-truth-free diagnostic with a measured operating characteristic (AUC 1.0 off the provably-undetectable coherent worst case), validate every identity against Monte-Carlo, and demonstrate both sides of the limit on real audio and scientific signals—the phenomenon persisting, fold-for-fold, for trained *nonlinear* and 2-D INRs. The classical uniform-lowpass fold is one special case; the structured-set fold law, the exact-fold characterization of silent aliasing, and the diagnostic’s operating characteristic are new.

Index Terms— Implicit neural representations, sampling theory, aliasing, generalized sampling, coordinate networks.

1. INTRODUCTION

Implicit neural representations parameterize a signal as a coordinate network $f_\theta : \mathbb{R}^d \rightarrow \mathbb{R}$ and are now a default continuous model for audio, images, volumes, and scientific fields [1, 2, 3]. A large literature engineers their *architectures* against spectral bias and aliasing—periodic activations, Fourier features, band-limited and anti-aliased variants [4, 5, 6, 7, 8, 9]—and a recent thread gives a *sampling theory*: Najaf and Ongie [10] and the super-resolution follow-up [11] prove that a signal which is a shallow INR is exactly recoverable from enough noiseless Fourier measurements, and sinc-activated INRs admit Shannon-style reconstruction guarantees [12]. These are *achievability* statements about realizable signals [13].

The gap. In practice the signal is not an INR: it is an arbitrary waveform observed at finitely many, often nonuniform, noisy

samples. The achievability theory is noiseless and realizable-signal by construction (misspecification appears in [10] only as an experiment); nothing above says when the task is possible, how the answer depends on the network’s capacity, or what failure looks like. We supply the converse. The key object is the INR’s *representable frequency set* Λ : for a Fourier-feature/periodic coordinate network the output is a combination of complex exponentials whose frequencies are integer combinations of the base feature frequencies—a set that grows with width and depth [14, 8]. A signal is representable only insofar as its spectrum lies in Λ .

Scope and contributions. Our theorems concern the analytically exact case: a fixed-feature INR whose output layer is linear over Λ . Its least-squares fit is generalized (nonharmonic) sampling on the finite dictionary Λ [15, 16]; that recast is the mechanism. On top of it we establish the INR-specific facts: (1) an achievability bound with a closed-form noise gain (Thm. 1); (2) a converse in which out-of-band energy folds onto Λ with an error floor, and an exact-fold characterization of *silent aliasing*—zero sample residual with full-energy reconstruction failure—whose target, for structured Λ , is unlike the classical lowpass law (Thm. 2); (3) an exact average-case error decomposition separating estimation variance from an aliasing-variance term that no sampling density uniformly removes (Thm. 3); and (4) a bandwidth-matched estimator, a ground-truth-free diagnostic with a measured ROC, Monte-Carlo-validated theory, and real-data demonstrations of both sides, which persist—fold-for-fold—for trained nonlinear and 2-D INRs.

Relation to prior analyses. When $\Lambda = \{-B, \dots, B\}$ and sampling is uniform, our converse reduces to classical aliasing (Rem. 1). Three adjacent lines must be distinguished. *Basis mismatch* [17] analyzes the same least-squares bias for a tone *near* a dictionary atom; our silent-aliasing regime is the complementary *far* regime, where a grid-coherent tone arbitrarily far from Λ folds with zero residual (Thm. 2(iii)). The *spectral power* of nonuniform sampling [18] predicts where out-of-band window appears in a periodogram; our fold target is its dictionary-projection analogue, and we pair it with achievability, noise, and a diagnostic. A *generalized aliasing decomposition* for least-squares/random-feature models [19] and an NTK-based lower bound from *missing* frequencies [20] both bound misspecification error; neither gives the structured fold law (where the energy *lands*), the exact-fold silent regime, or the detection question that we address. Empirically, trained SIRENs have been observed to plant off-grid aliases [21]; we supply the theory and the diagnostic for that observation.

2. SIGNAL MODEL AND REPRESENTABLE BAND

Let $f : [0, 1) \rightarrow \mathbb{R}$ be observed at samples $t_1, \dots, t_N$ (possibly nonuniform) as $y_j = f(t_j) + \varepsilon_j$. A fixed-feature INR with feature set inducing $\Lambda = \{\omega_1, \dots, \omega_m\} \subset \mathbb{R}$ has model class $\mathcal{M}_\Lambda = \{\sum_k c_k e_{\omega_k}\}$ where $e_\omega(x) = e^{i2\pi\omega x}$; deeper/periodic networks en-

large Λ to integer combinations of base frequencies [14]. Define the *synthesis matrix* $\Phi \in \mathbb{C}^{N \times m}$, $\Phi_{jk} = e_{\omega_k}(t_j)$; fitting the INR is the least-squares problem $\hat{\mathbf{c}} = \arg \min_{\mathbf{c}} \|\mathbf{y} - \Phi \mathbf{c}\|^2$ (the released code optionally adds a negligible ridge, 10^{-8} – 10^{-6} in the experiments). Write $f = f_{\text{in}} + f_{\text{out}}$ where $f_{\text{in}} = \sum_k c_k^* e_{\omega_k} \in \mathcal{M}_\Lambda$ and $f_{\text{out}} = \sum_l a_l e_{\nu_l}$ is a finite exponential sum on frequencies $\nu_l \notin \Lambda$ (synthesis matrix Φ_{out}), so $\mathbf{y} = \Phi \mathbf{c}^* + \mathbf{g} + \varepsilon$ with $g_j = f_{\text{out}}(t_j)$. The lower *frame bound* is $A = \sigma_{\min}(\Phi)^2$; $A > 0$ iff Φ has full column rank ($m \leq N$, generic positions), i.e. the samples determine $\mathcal{M}_\Lambda$.

3. FUNDAMENTAL LIMITS

Theorem 1 (Achievability). *If $f_{\text{out}} = 0$ and Φ has full column rank ($\sigma_{\min}(\Phi) > 0$), then $\|\hat{\mathbf{c}} - \mathbf{c}^*\| \leq \kappa \|\varepsilon\|$ with noise gain $\kappa = \|(\Phi^* \Phi)^{-1} \Phi^*\|_2 = 1/\sigma_{\min}(\Phi)$, and for white noise of variance σ^2 , $\mathbb{E}\|\hat{\mathbf{c}} - \mathbf{c}^*\|^2 = \sigma^2 \text{tr}((\Phi^* \Phi)^{-1})$.*

Full column rank needs $N \geq m = |\Lambda|$; for the special case $\Lambda = \{-B, \dots, B\}$ this is consistent with the Landau density $N \gtrsim 2B$ [22], but for a sparse structured Λ the binding requirement is $N \geq |\Lambda|$, not the band extent. Rank alone prices nothing: κ can be arbitrarily large for clustered samples—the bound is achievability at the price κ , which generalized sampling controls via density conditions [15].

Theorem 2 (Converse and aliasing). *Assume $\sigma_{\min}(\Phi) > 0$ and let $P = \Phi(\Phi^* \Phi)^{-1} \Phi^*$, $\mathbf{M} = (\Phi^* \Phi)^{-1} \Phi^*$. With $f_{\text{out}} \neq 0$ the estimate is biased, $\hat{\mathbf{c}} = \mathbf{c}^* + \Delta \mathbf{c}$, $\Delta \mathbf{c} = \mathbf{M} \mathbf{g}$, and: (i) no INR in $\mathcal{M}_\Lambda$ fits the noiseless samples better than $\min_{\mathbf{c}} \|\mathbf{y} - \Phi \mathbf{c}\| = \|(\mathbf{I} - P) \mathbf{g}\|$; (ii) for a tone $a e_{\nu}$ with sample vector ϕ_ν , the residual is $|a| \|\phi_\nu\| \sin \theta_\nu$ (θ_ν the principal angle between ϕ_ν and range Φ) while the spectral bias is $\|\Delta \mathbf{c}\| = |a| \|\mathbf{M} \phi_\nu\|$; (iii) (silent aliasing) if $\Lambda \subset \mathbb{Z}$, the samples lie on a rate- Q grid ($t_j \in \{0, \dots, Q-1\}/Q$), and $\nu \in \mathbb{Z} \setminus \Lambda$ satisfies $\nu \equiv \omega_k \pmod{Q}$, then $\phi_\nu = \phi_{\omega_k}$ exactly: the residual is zero while $\|\Delta \mathbf{c}\| = |a|$, and every $g \in \mathcal{M}_\Lambda$ has $\|g - f\|_{L^2[0,1]} \geq \|f_{\text{out}}\|_{L^2}$ —full-energy reconstruction failure, invisible in-sample, at any N and with ν arbitrarily far from Λ .*

We call such a fold *coherent* with the sampling; in general, coherence means $\|\mathbf{M} \phi_\nu\|$ stays bounded away from zero as the density grows. The *fold target* is defined as ω_{k^*} , $k^* = \arg \max_k |\Delta c_k|$; for generic nonuniform samples it is the dictionary analogue of the peak of the sampled spectral window familiar from Lomb–Scargle analysis [18]. The near-collision regime $\nu \rightarrow \omega_k$ (where the residual also vanishes but so does the reconstruction error) is the basis-mismatch sensitivity of [17], governed by super-resolution conditioning [23]; (iii) is its harmful far counterpart.

Remark 1. *For uniform $t_j = j/N$ and Λ an integer lowpass band, the fold target is the unique element of Λ congruent to $\nu \pmod{N}$ (and $\Delta \mathbf{c} = \mathbf{0}$ if the residue class misses Λ): classical aliasing [24] is the coherent special case $Q = N$. For structured Λ and grid-subset or nonuniform samples the fold target is non-classical (Fig. 1).*

Theorem 3 (Average-case decomposition). *Let the out-of-band coefficients a_l be zero-mean, uncorrelated, with $\mathbb{E}|a_l|^2 = p_l$, independent of the noise, and $\mathbb{E}[\varepsilon \varepsilon^*] = \sigma^2 \mathbf{I}$. Then the mean-squared error of the least-squares estimator is exactly*

$$\mathbb{E}\|\hat{\mathbf{c}} - \mathbf{c}^*\|^2 = \underbrace{\sigma^2 \text{tr}((\Phi^* \Phi)^{-1})}_{\text{estimation variance}} + \underbrace{\text{tr}(\mathbf{M} \Sigma_g \mathbf{M}^*)}_{\text{aliasing variance}}, \quad (1)$$

with $\Sigma_g = \Phi_{\text{out}} \text{diag}(\mathbf{p}) \Phi_{\text{out}}^*$.

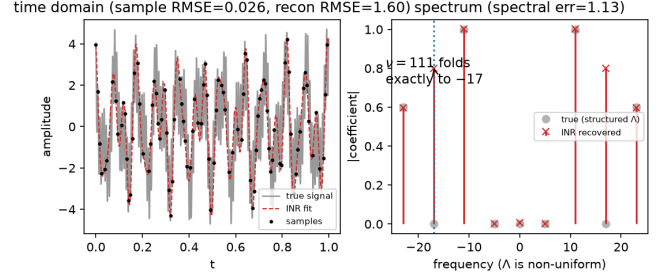


Fig. 1. *Silent aliasing onto a structured $\Lambda = \{0, \pm 5, \pm 11, \pm 17, \pm 23\}$ ($m=9$): the exact-fold regime of Thm. 2(iii). $N=96$ nonuniform samples (a random subset of a rate- $Q=128$ grid, $N \gg m$), noise $\sigma=0.03$. The INR fits the samples at the noise floor (RMSE 0.026; 0.028 ± 0.002 over 20 draws) yet its reconstruction error is 1.60 and its spectrum carries the full tone energy (bias 1.13): the tone $\nu=111$ folds *exactly* onto the non-classical atom -17 in every draw, while the true tones ($\pm 11, \pm 23$) are kept.*

Equation (1) is a fixed- N identity. The estimation term decays like $\sigma^2 m/N$ for well-conditioned sampling, but the aliasing term is not *uniformly* removable: at every density some tone folds coherently ($\|\mathbf{M} \phi_\nu\| = 1$, Thm. 2(iii)) and keeps it at full power. For a *fixed* tone the term converges to its window-leakage energy onto $\mathcal{M}_\Lambda$: exactly zero for integer-disjoint tones under uniform sampling ($O(m/N)$ in expectation under i.i.d. sampling), but the positive constant $\sum_k \text{sinc}^2(\nu - \omega_k)$ for non-integer ν . The proofs are elementary least-squares/projection identities; each statement—including every asymptotic regime above—is asserted against Monte-Carlo in the released test suite.

4. MATCHED ESTIMATOR AND DIAGNOSTIC

Bandwidth matching. Estimating the essential band from the samples via a nonuniform periodogram and choosing Λ to cover it places the signal in the Thm. 1 regime; this is the estimator we use. **Diagnostic.** Because a fold is invisible in-sample (Thm. 2(ii)–(iii)), we test band-limitation to Λ by fitting an enlarged dictionary $\Lambda \cup \Lambda_{\text{ring}}$ (Λ_{ring} a ring of candidate frequencies outside Λ) and measuring the recovered energy fraction on the ring. The test is valid only in the overdetermined regime $N > |\Lambda \cup \Lambda_{\text{ring}}|$ (an underdetermined ridge fit spreads energy everywhere and flags every signal), and its threshold should be null-calibrated; Sec. 5 measures its full operating characteristic. By Thm. 2(iii) an *exactly* coherent grid fold is information-theoretically undetectable from the samples—the diagnostic’s measured collapse there is the converse made visible, not a defect of the test.

5. EXPERIMENTS

Every identity is validated against Monte-Carlo (frame bounds, κ , aliasing floor, exact and near-collision fold regimes, all three Thm. 3 asymptotic regimes, and the joint decomposition (1)). Fig. 1 shows the structured silent-aliasing regime of Thm. 2(iii): the INR fits the samples at the noise floor yet the recovered spectrum carries the full tone energy on the exact non-classical fold atom. Fig. 2 shows the phase transition with the exact theory overlaid.

Diagnostic ROC. Over 500 draws per condition (structured Λ as

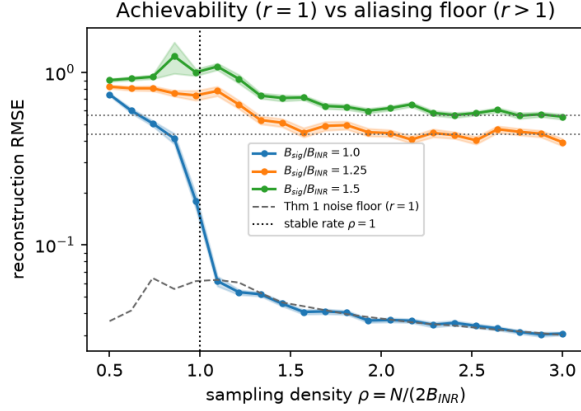


Fig. 2. Achievability vs. converse (Thm. 1–2), 40-trial Monte-Carlo (bands: ± 1 s.e.), $r = B_{\text{sig}}/B_{\text{INR}}$, sampling density $\rho = N/(2B_{\text{INR}})$. In-band ($r=1$) error is large in the rank-deficient regime $\rho < 1$, drops sharply at the stable rate $\rho=1$, and then tracks the exact Thm. 1 noise floor (dashed); out-of-band ($r>1$) error plateaus at the predicted $\|f_{\text{out}}\|$ levels (dotted) that no density removes.

in Fig. 1, ring $[-30, 30] \setminus \Lambda$, $N \in \{80, 150\}$ jittered samples, tone amplitudes $0.25\text{--}1\times$ the in-band RMS, on- and off-grid ν , the ring diagnostic separates perfectly (AUC 1.000; TPR 1.000 at the null-calibrated threshold with FPR 0.05; the fixed 0.1 threshold has zero false positives but misses the weakest tones). On the exact coherent grid fold of Fig. 1 it collapses to below-chance (AUC 0.20)—the provably undetectable worst case—and in the underdetermined regime it flags 100% of null signals, confirming the validity requirement of Sec. 4 (released `diagnostic_roc`).

Real data, both sides (Fig. 3; mean $\pm$ s.d. over 10 subsample/noise draws). Sweeping the INR bandwidth B_{INR} on a real signal and reading the reconstruction-error knee gives the empirical learned-Nyquist rate. Mauna Loa CO_2 (NOAA; $N=414$ of $M=820$, nonuniform), after linear detrending, is bandlimited (annual cycle), giving a sharp knee at its spectral peak: a narrow INR aliases it, a matched INR recovers it (RMSE $0.42 \pm 0.00 \rightarrow 0.17 \pm 0.03$). Broadband speech (Open Speech Repository; $N=1400$ of $M=2000$) shows the converse side: no width approaches the 0.03 noise floor—narrow bands sit on a truncation-plus-aliasing floor (RMSE ≥ 1.02) and widths approaching the sample budget explode from the Thm. 1 noise gain ($\kappa \rightarrow \infty$ as $m \rightarrow N$; RMSE 682 ± 386 at $B=650$). Smoothed sunspots (SILSO; the smoothing is a documented lowpass proxy) behave like CO_2 ($0.92 \pm 0.01 \rightarrow 0.41 \pm 0.18$); the raw series behaves broadband. The phenomenon persists, fold-for-fold, for trained *nonlinear* INRs: training identical Fourier-feature networks with and without an out-of-hull tone and taking the difference spectrum isolates where the trained net puts the tone’s energy—it matches the Hermitian-pair linear bias pattern bin-for-bin (correlation 0.99/0.91/0.63 over three sampling draws; measured peak equal to the predicted fold target in two, and to a near-tied atom within 3% in the third), while a bandwidth-covering trained network recovers real CO_2 slightly better than the linear estimator (0.107 vs. 0.143). Because the Gaussian feature scale is a *soft* band edge (features have tails and the MLP generates harmonics), the trained-INR persistence claim rests on this out-of-hull-tone ablation, not on a scale sweep. Finally, a 2-D INR on sparse image pixels (Fig. 4) reproduces the limit: an

over-scaled band fits the observed pixels to 51 dB while the full field is moiré-corrupted (16 dB).

6. CONCLUSION

Fixed-feature INRs alias, and provably so: they represent only a structured band Λ , out-of-band energy folds onto Λ with an error floor, and for coherent folds the corruption is total yet *exactly* invisible in the sample fit. The limit is sharp, matched by a bandwidth-selecting estimator, detectable from data everywhere except the provably-undetectable coherent worst case, and visible on real signals, persisting fold-for-fold for trained nonlinear and 2-D INRs. Extending the converse to adaptive-frequency Λ (where the network can move atoms onto the signal) and to the exact training dynamics are the natural next steps.

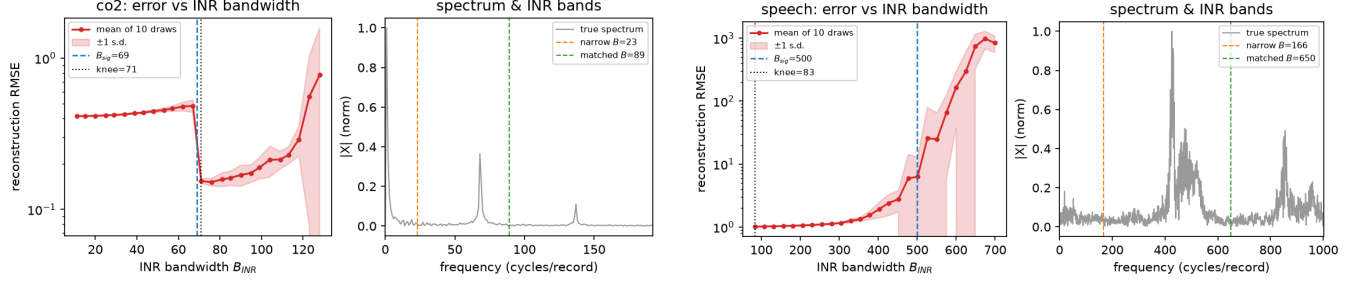


Fig. 3. Both sides of the limit on real data (curves: mean $\pm$ 1 s.d. over 10 subsample/noise draws; dashed line B_{sig} = essential bandwidth, dotted line = error knee). *Left* (CO_2 , recoverable): the knee sits at the annual-cycle frequency; a narrow INR aliases, a matched INR recovers (the mild rise past the knee is the onset of the noise gain). *Right* (speech, converse): the error never approaches the noise floor at any width—a truncation-plus-aliasing floor for narrow bands, and a noise-gain blow-up ($\kappa = 1/\sigma_{\min} \rightarrow \infty$) as the width approaches the sample budget.

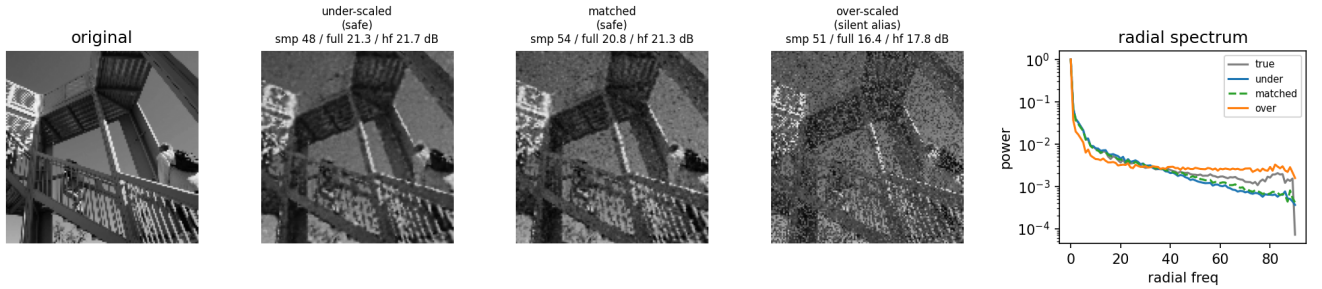


Fig. 4. The limit generalizes to a trained *nonlinear* 2-D INR fit to sparse image pixels (panel titles: “smp” = PSNR on the observed pixels, “full” = full-field PSNR, “hf” = PSNR restricted to the upper radial-frequency band). An over-scaled representable band silently aliases: it fits the observed pixels (51 dB) but the full field is moiré-corrupted (16 dB) and its radial spectrum carries spurious high-frequency energy; a band matched to (or below) the sampling is safe—under- and matched-band fits are statistically indistinguishable here; the contrast that matters is with the over-scaled band.

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